PLANE WAVE PARAMETERS (LOSSLESS)

Angular frequency *ω: rad/s; f: Hz*

$$\omega = 2\pi f \qquad f = \frac{\omega}{2\pi}$$

Phase velocity *m/s; nonmagnetic: $\mu_r = 1$*

$$u_p = \frac{c}{\sqrt{\varepsilon_r}} \qquad c = 3\times 10^8 \text{ m/s}$$

Wavenumber k *rad/m*

$$k = \frac{\omega}{u_p} = \frac{2\pi}{\lambda} = \frac{\omega\sqrt{\varepsilon_r}}{c}$$

Wavelength & impedance *m & Ω*

$$\lambda = \frac{u_p}{f} = \frac{2\pi}{k}\ (\text{m}) \qquad \eta = \frac{\eta_0}{\sqrt{\varepsilon_r}}\ (\Omega),\ \eta_0 = 377\,\Omega$$

Order: $\omega \to u_p \to k \to \lambda \to \eta$. Base: $\mathbf{E} = E_0\cos(\omega t \pm kx + \phi_0)\hat{e}$.

LOSSLESS PLANE WAVE — E AND H

General E

$$\mathbf{E} = E_0\cos(\omega t - k\,\hat{k}\cdot\mathbf{r} + \phi_0)\,\hat{e}$$

$\hat{k}$ = direction of travel (unit vector), e.g. $+y \Rightarrow \hat{k} = \hat{y}$.
k = wavenumber (rad/m): how fast phase changes in space; $k = \omega\sqrt{\varepsilon_r}/c = 2\pi/\lambda$.
ϕ_0 = initial phase constant (rad): shift set by given condition at known t, position.
$\hat{e}$ = polarization direction of $\mathbf{E}$ (unit vector), e.g. E along $x \Rightarrow \hat{e} = \hat{x}$.
$\mathbf{r} = x\hat{x} + y\hat{y} + z\hat{z}$, so $\hat{k}\cdot\mathbf{r}$ picks the axis, e.g. $\hat{k} = \hat{y} \Rightarrow \hat{k}\cdot\mathbf{r} = y$.

H from E *$\hat{k}$, $\hat{E}$, $\hat{H}$ form right-hand set*

$$\mathbf{H} = \frac{1}{\eta}\,(\hat{k}\times\mathbf{E})$$

Find ϕ_0 *given $E = E_1$ at $t = 0$, position r_0*

$$E_0\cos(-kr_0 + \phi_0) = E_1 \ \Rightarrow\ \phi_0 = \cos^{-1}\!\left(\frac{E_1}{E_0}\right) + kr_0$$

$\cos^{-1}$ gives $[0,\pi]$; take ϕ_0 in that range.

DIRECTION OF PROPAGATION

Arg. contains	$\hat{k}$	Arg. contains	$\hat{k}$
$\omega t - kz$	$+\hat{z}$	$\omega t + kz$	$-\hat{z}$
$\omega t - ky$	$+\hat{y}$	$\omega t + ky$	$-\hat{y}$
$\omega t - kx$	$+\hat{x}$	$\omega t + kx$	$-\hat{x}$

Minus before k-axis $\Rightarrow$ + direction. Plus $\Rightarrow$ − direction.

$\hat{k}\times\hat{E}$ — FIND $\hat{H}$ DIRECTION

$\hat{k}$	$\hat{E}$	$\hat{H}$	$\hat{k}$	$\hat{E}$	$\hat{H}$
$+\hat{z}$	$\hat{x}$	$+\hat{y}$	$+\hat{z}$	$\hat{y}$	$-\hat{x}$
$+\hat{y}$	$\hat{x}$	$-\hat{z}$	$+\hat{y}$	$\hat{z}$	$+\hat{x}$
$+\hat{x}$	$\hat{y}$	$+\hat{z}$	$+\hat{x}$	$\hat{z}$	$-\hat{y}$
$-\hat{x}$	$\hat{y}$	$-\hat{z}$	$-\hat{x}$	$\hat{z}$	$+\hat{y}$
$-\hat{y}$	$\hat{x}$	$+\hat{z}$	$-\hat{z}$	$\hat{x}$	$-\hat{y}$

Cyclic: $\hat{x}\times\hat{y} = \hat{z}$, $\hat{y}\times\hat{z} = \hat{x}$, $\hat{z}\times\hat{x} = \hat{y}$. Reverse $\Rightarrow$ flip sign.

TRIG PHASE SHIFTS *($\theta \equiv \omega t - kz$)*

Input	Result	Input	Result
$\sin(\theta+\pi/2)$	$\cos\theta$	$\cos(\theta+\pi/2)$	$-\sin\theta$
$\sin(\theta-\pi/2)$	$-\cos\theta$	$\cos(\theta-\pi/2)$	$\sin\theta$
$\sin(\theta\pm\pi)$	$-\sin\theta$	$\cos(\theta\pm\pi)$	$-\cos\theta$

POLARIZATION — GENERAL WAVE

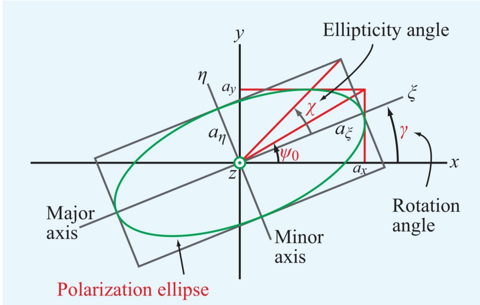

Figure 7-11 Polarization ellipse in the x–y plane with the wave traveling in the z direction (out of the page).

$$\mathbf{E} = a_x\cos(\omega t - kz)\hat{x} + a_y\cos(\omega t - kz + \delta)\hat{y}$$

a_x, a_y: bounding box half-widths (V/m); δ: phase diff (rad)
$\delta = 0$ or $\pi \Rightarrow$ ***linear*** *pol. ($\chi = 0$, no ellipse)* **Auxiliary angle** $\psi_0 = \arctan(a_y/a_x) \in (0, \pi/2)$ **Rotation angle** γ **(major axis from $\hat{x}$)** $\gamma \in [0, \pi)$

$$\tan(2\gamma) = \tan(2\psi_0)\cos\delta$$

If sign(γ) $\neq$sign(δ): $\gamma \leftarrow \gamma \pm \frac{\pi}{2}$. **Ellipticity angle** χ $\chi \in [-\pi/4, \pi/4]$

$$\sin(2\chi) = \sin(2\psi_0)\sin\delta$$

Axial ratio $R = 1/\tan|\chi|$ *major axis = $R\times$ minor; $R \ge 1$*

χ SHAPE & HANDEDNESS

χ describes the shape of the ellipse and rotation direction.

χ (rad)	Polarization Type	Handedness
0	Linear	None
$(0,\ \pi/4)$	RHEP	RH (CCW)
$\pi/4$	RHC	RH (CCW)
$-\pi/4$	LHC	LH (CW)
$(-\pi/4,\ 0)$	LHEP	LH (CW)

QUADRANT RULE FOR $\tan(2\gamma)$

$N = \sin(2\psi_0)\cos\delta$, $\quad D = \cos(2\psi_0)$:

N	D	2γ (deg)	2γ (rad)
+	+	$\arctan(N/D)$	$\arctan(N/D)$
+	−	$180^\circ - \arctan\lvert N/D\rvert$	$\pi - \arctan\lvert N/D\rvert$
−	−	$180^\circ + \arctan\lvert N/D\rvert$	$\pi + \arctan\lvert N/D\rvert$
−	+	$360^\circ - \arctan\lvert N/D\rvert$	$2\pi - \arctan\lvert N/D\rvert$

$\gamma = (2\gamma)/2$. $\quad D = 0 \Rightarrow 2\gamma = \frac{\pi}{2} \Rightarrow \gamma = \frac{\pi}{4}$.
Convert: $\theta_{rad} = \theta_\circ \times \frac{\pi}{180}$; $\theta_\circ = \theta_{rad}\times\frac{180}{\pi}$.

PHASOR ↔ TIME DOMAIN

Phasor coeff	Time domain (add $e^{-\alpha z}$)
A (real)	$A\cos(\omega t - \beta z)$
$-jA$	$A\sin(\omega t - \beta z)$
$+jA$	$-A\sin(\omega t - \beta z)$
$Ae^{j\phi}$	$A\cos(\omega t - \beta z + \phi)$
$A + jB$	$\lvert Z\rvert\cos(\omega t - \beta z + \theta)$

$|Z| = \sqrt{A^2 + B^2}$, $\theta = \arctan(B/A)$ (check quadrant)

LOSSY MEDIA — CLASSIFICATION

Loss tangent

$$\frac{\varepsilon''}{\varepsilon'} = \frac{\sigma}{\omega\varepsilon_r\varepsilon_0}$$

$\varepsilon''/\varepsilon'$	Type	Use
> 100	Good conductor	GC approx
0.01 to 100	Quasi-conductor	Exact eqs
< 0.01	Low-loss dielectric	LD approx

Non-magnetic ($\mu_r = 1$): $\mu = \mu_0$. $\varepsilon' = \varepsilon_r\varepsilon_0$ ($\varepsilon_r = 1$: vacuum/air).

EXACT FORMULAS (QUASI-CONDUCTOR)

Let $X = \sqrt{1 + (\sigma/\omega\varepsilon)^2} = \sqrt{1 + (\varepsilon''/\varepsilon')^2}$, $\varepsilon'' = \sigma/\omega$, $\varepsilon = \varepsilon_r\varepsilon_0$, $\mu = \mu_0$
α and β

$$\alpha = \omega\sqrt{\frac{\mu\varepsilon}{2}}\sqrt{X-1}\ (\text{Np/m}), \quad \beta = \omega\sqrt{\frac{\mu\varepsilon}{2}}\sqrt{X+1}\ (\text{rad/m})$$

$$\lambda = \frac{2\pi}{\beta}\ (\text{m}), \quad u_p = \frac{\omega}{\beta}\ (\text{m/s})$$

$$\eta_c = \frac{\eta}{X^{1/2}}\angle\theta_\eta\ (\Omega), \quad \theta_\eta = \tfrac{1}{2}\arctan\tfrac{\sigma}{\omega\varepsilon}\ (\text{deg})$$

where $\eta = \eta_0/\sqrt{\varepsilon_r}$.

GOOD CONDUCTOR ($\sigma/\omega\varepsilon \gg 100$)

$$\alpha \approx \omega\sqrt{\frac{\mu\varepsilon''}{2}} = \omega\sqrt{\frac{\mu\sigma}{2\omega}} = \sqrt{\pi f\mu\sigma} \quad (\text{Np/m})$$

$$\beta \approx \alpha \approx \sqrt{\pi f\mu\sigma} \quad (\text{rad/m})$$

$$u_p = \omega/\beta\ (\text{m/s}), \quad \lambda = 2\pi/\beta\ (\text{m})$$

$$\eta_c \approx \sqrt{j\frac{\mu}{\varepsilon''}} = (1+j)\sqrt{\frac{\pi f\mu}{\sigma}} \approx (1+j)\frac{\alpha}{\sigma} \quad (\Omega)$$

LOW-LOSS DIELECTRIC ($\sigma/\omega\varepsilon \ll 0.01$)

$$\alpha \approx \frac{\omega\varepsilon''}{2}\sqrt{\frac{\mu}{\varepsilon'}} = \frac{\sigma}{2}\sqrt{\frac{\mu}{\varepsilon'}} = \frac{\sigma\eta}{2}\ (\text{Np/m})$$

$$\beta \approx \omega\sqrt{\mu\varepsilon'} = \frac{\omega\sqrt{\varepsilon_r}}{c}\ (\text{rad/m}), \quad u_p \approx \frac{c}{\sqrt{\varepsilon_r}}\ (\text{m/s})$$

$$\eta_c \approx \sqrt{\frac{\mu}{\varepsilon'}} = \eta\ (\Omega,\ \angle \approx 0^\circ)$$

$\tilde{H}$ PHASOR → $\tilde{E}$ IN LOSSY MEDIA

Step 1 — read α, β

$$\tilde{\mathbf{H}} \propto e^{-\alpha y}e^{-j\beta y} \ \Rightarrow\ \gamma = \alpha + j\beta$$

Step 2 — η_c (nonmagnetic: $\mu_r = 1$)

$$\eta_c = \frac{j\omega\mu_0}{\gamma} = \frac{\omega\mu_0(\beta + j\alpha)}{\alpha^2 + \beta^2}$$

Step 3 — find $\tilde{\mathbf{E}}$

$$\tilde{\mathbf{E}} = -\eta_c\,(\hat{k}\times\tilde{\mathbf{H}})$$

Step 4 — time domain

$$Ae^{j\phi}e^{-\alpha y} \xrightarrow{\text{Re}\{\cdot e^{j\omega t}\}} Ae^{-\alpha y}\cos(\omega t - \beta y + \phi)$$

Convert each component $A + jB \to |Z|\angle\theta$ first.

AVERAGE POWER DENSITY (POYNTING)

General *W/m^2*

$$\mathbf{S}_{\text{av}} = \tfrac{1}{2}\,\text{Re}\{\tilde{\mathbf{E}}\times\tilde{\mathbf{H}}^*\}$$

Lossless shortcut $|E_0|^2 = a_x^2 + a_y^2 + a_z^2$

$$\mathbf{S}_{\text{av}} = \frac{|E_0|^2}{2\eta}\,\hat{k}$$

Direction = $\hat{k}$. If wave goes in $-\hat{x}$, then $\mathbf{S}_{av}$ is a negative x-component.

WAVE PARAMS

$\omega = 2\pi f$ — ang. freq (rad/s)
$k = \omega\sqrt{\varepsilon_r}/c$ — wavenumber (rad/m)
$\lambda = 2\pi/k$ — wavelength (m)
$u_p = c/\sqrt{\varepsilon_r}$ — phase vel. (m/s)
$\eta = \eta_0/\sqrt{\varepsilon_r}$ — impedance (Ω)
$c = 3\times 10^8$ m/s
$\eta_0 = 377\,\Omega$ (free space)

LOSSLESS MEDIA

$\sigma = 0$ (vacuum, air, ideal dielectric)
$\varepsilon'' = \sigma/\omega = 0 \Rightarrow \varepsilon = \varepsilon'$ (real)
$\alpha = 0$; no attenuation
$\eta = \eta_0/\sqrt{\varepsilon_r}$ real (Ω)
$\eta_0 = 377\,\Omega$ (free space)
E and **H** in phase ($\theta_\eta = 0$)

LOSSY MEDIA

α — atten. const (Np/m)
β — phase const (rad/m)
$\gamma = \alpha + j\beta$ — prop. const (m^{-1})
η_c — complex impedance (Ω)
$\varepsilon' = \varepsilon_r\varepsilon_0$; $\varepsilon'' = \sigma/\omega$ (F/m)
μ_r — rel. permeability ($\mu_r = 1$: nonmagnetic)
$\mu = \mu_r\mu_0$; $\mu_0 = 4\pi\times 10^{-7}$ H/m
$\delta_s = 1/\alpha$ — skin depth (m)

CIRCULAR WAVES

$\tilde{\mathbf{E}}_R = a_R(\hat{x} + e^{-j\pi/2}\hat{y})e^{-jkz} = a_R(\hat{x} - j\hat{y})e^{-jkz}$
$\tilde{\mathbf{E}}_L = a_L(\hat{x} + e^{+j\pi/2}\hat{y})e^{-jkz} = a_L(\hat{x} + j\hat{y})e^{-jkz}$
a_R, a_L, a_x in V/m; k in rad/m
Linear: $a_R = a_L = a_x/2$
$-j = e^{-j\pi/2}$: cos→sin
$+j = e^{+j\pi/2}$: cos→ −sin

USEFUL NUMBERS

$\sqrt{2.56} = 1.6$; $\sqrt{9} = 3$
$\sqrt{2.5} \approx 1.581$; $\sqrt{2} \approx 1.414$
$\arccos(2/3) \approx 48.2^\circ = 0.84$ rad
$\arctan(4/3) \approx 53.1^\circ = 0.927$ rad
$\arctan(3/4) \approx 36.9^\circ = 0.644$ rad
$\arctan(24/7) \approx 73.7^\circ = 1.287$ rad